Employee Management System

A console-based Employee Management System developed in C++ using Object-Oriented Programming (OOP) principles. This application allows users to manage employee records efficiently through a menu-driven interface.

Overview

The Employee Management System is designed to perform basic employee record management operations such as adding, updating, removing, and viewing employee information. The project demonstrates the practical implementation of OOP concepts and dynamic data storage using vectors.

Features

- Add new employees
- Remove existing employees
- Update employee details
- View all employee records
- User-friendly menu-driven interface
- Dynamic data storage using STL vectors

Technologies Used

- C++
- Object-Oriented Programming (OOP)
- Standard Template Library (STL)
- Vectors
- Console Application Development

Functionalities

Add Employee

Allows the user to add a new employee by entering:

- Employee Name
- Employee Role
- Employee Salary

Remove Employee

Removes an employee record using the employee's name.

Update Employee

Updates:

- Employee Role
- Employee Salary

View Employees

Displays a formatted list of all employee records.

Project Structure

```
Employee-Management-System/
|
├── main.cpp
├── README.md
├── Project_Report.pdf
└── Screenshots/
```

Getting Started

Clone the Repository

```
git clone https://github.com/abdulrehman2301/Employee-Management-System.git
```

Navigate to Project Directory

```
cd Employee-Management-System
```

Compile the Program

```
g++ main.cpp -o ems
```

Run the Program

```
./ems
```

Sample Menu

```
--- Employee Management System ---

1. Add Employee
2. Remove Employee
3. Update Employee
4. View Employees
```

5. Exit

Learning Outcomes

This project demonstrates:

- Class and Object implementation
- Constructors and member functions
- Encapsulation
- Vector-based data management
- Menu-driven application development
- Input and output handling in C++

Academic Information

Course: Object Oriented Paradigm
Student: Muhammad Abdul Rehman

License

This project was developed for academic and educational purposes.